

Analytically Solvable Models of the Electromagnetic Response of a Spherical Earth to External Current Systems

1 Motivation

Consider a spherical Earth of radius a surrounded by ionospheric current systems. A natural idealization of an ionospheric source is a circular current loop in the equatorial plane. This document develops that model, but its main deliverable is Section 5: a fully closed-form, single-spherical-harmonic solution that can be evaluated to machine precision and used directly to validate a numerical solver, with no series truncation and no dependence on the (more delicate) loop-multipole expansion.

2 Model I: Homogeneous Conducting Earth — General Framework

Let the Earth be a homogeneous sphere of radius a , conductivity σ , permeability μ , in vacuum ($\sigma = 0$, permeability μ_0) outside. An external source (loop, or later a surface current shell) drives fields with time dependence $e^{i\omega t}$.

Inside the Earth (quasi-static, displacement current neglected):

$$\nabla \times \mathbf{H} = \sigma \mathbf{E}, \quad (1)$$

$$\nabla \times \mathbf{E} = -i\omega \mathbf{B}, \quad (2)$$

$$\mathbf{B} = \mu \mathbf{H}. \quad (3)$$

Taking the curl of (1) and using (2) together with $\nabla \cdot \mathbf{H} = 0$ gives the vector Helmholtz equation

$$\nabla^2 \mathbf{H} = i\omega\mu\sigma \mathbf{H}. \quad (4)$$

Remark 1 (Sign convention for the radial solutions). *Equation (4) is often paired in the literature with the statement “the radial solutions are spherical Bessel functions $j_\ell(kr)$, $y_\ell(kr)$ with $k^2 = i\omega\mu\sigma$.” This is only true if k^2 is defined with the opposite sign. Separating (4) for an axisymmetric poloidal field (Section 5) gives a radial ODE of the form*

$$R'' - \frac{\ell(\ell+1)}{r^2} R = i\omega\mu\sigma R.$$

The function $r j_\ell(qr)$ satisfies $R'' - \ell(\ell+1)R/r^2 + q^2 R = 0$ (ordinary Helmholtz form, verified symbolically below), so matching to the equation above requires $q^2 = -i\omega\mu\sigma$. Throughout this document we therefore define

$$k^2 = -i\omega\mu\sigma, \quad (5)$$

so that the radial solutions are literally $j_\ell(kr)$, $y_\ell(kr)$, and $k = (1 - i)/\delta$ in terms of the skin depth $\delta = \sqrt{2/(\omega\mu\sigma)}$ (with $\omega > 0$). This is the opposite sign from the original draft's $k^2 = i\omega\mu\sigma$; that draft's formulas $j_\ell(kr)$ were only valid if one silently used this convention. Any numerical benchmark must fix this sign before comparing phases — it is the single most common source of a spurious 90° phase offset in this kind of code.

The magnetostatic limit $\omega \rightarrow 0$ (Section 5.5) is recovered by $k \rightarrow 0$, where $j_\ell(kr) \rightarrow (kr)^\ell/(2\ell + 1)!!$, reducing the radial solutions to ordinary powers r^ℓ .

3 Loop Source: Structure of the Solution

A circular loop of radius $R > a$ in the equatorial plane, carrying $I(t) = I_0 e^{i\omega t}$, is the physically natural ionospheric idealization. Its scalar/vector potential outside the Earth is a superposition of the loop's own field and a scattered field:

$$\Phi_{\text{out}}(r, \theta) = \Phi_{\text{loop}}(r, \theta) + \sum_{\ell} b_{\ell} r^{-\ell-1} P_{\ell}(\cos \theta), \quad a < r, r \neq R. \quad (6)$$

Remark 2 (Φ_{loop} is piecewise, not a single series). Φ_{loop} itself is only single-valued as one Legendre series for $r < R$ (regular, $\propto r^\ell$ terms) and a different series for $r > R$ (decaying, $\propto r^{-\ell-1}$ terms); the two only agree in that their sum reproduces the loop's field on each side. Since the Earth's surface $r = a$ satisfies $a < R$, only the interior branch of Φ_{loop} is ever evaluated in the boundary conditions at $r = a$, so (6) is safe to use there. It should not be read as a single formula valid uniformly for all $r > a$ (it has a discontinuity in form, not in value, at $r = R$).

Remark 3 (Boundary condition at $r = a$). The correct pair of conditions at a source-free interface is continuity of Φ_m itself (equivalent to continuity of the tangential H , up to an additive constant that is fixed by regularity/decay at ∞) and continuity of $\mu \partial \Phi_m / \partial r$ (normal B). Stating the first condition as $\partial \Phi_{\text{in}} / \partial \theta = \partial \Phi_{\text{out}} / \partial \theta$ is equivalent for every $\ell \geq 1$ but silently drops the $\ell = 0$ (monopole) equation; this is harmless here only because a current loop has no monopole moment ($a_0 = b_0 = 0$ by construction).

Getting Φ_{loop} 's Legendre coefficients right (elliptic-integral / associated-Legendre-function algebra) is genuinely the hard part of this model and a second source of numerical error on top of whatever the solver itself contributes. For validating a numerical code, Section 5 below is the more useful tool: it removes the loop entirely and is exact by construction.

4 Layered Earth: General Pattern

For N concentric homogeneous shells plus the exterior vacuum, each harmonic degree ℓ decouples (Legendre/associated-Legendre orthogonality) into an independent linear system of size $2N \times 2N$: two unknown radial coefficients per interior region, continuity of Φ_m (or of r -times-flux-function, in the diffusive case) and of the normal- B combination $\mu \partial \Phi_m / \partial r$ at each of the N interfaces. Section 5 works this out explicitly and *verifies it symbolically* for the simplest nontrivial case (one conducting sphere, one source shell, vacuum in between and outside) — the pattern for additional shells is a direct extension of the same matching procedure.

5 The Benchmark Problem: A Single Spherical Harmonic

5.1 Why a surface current shell instead of a loop

A circular loop excites *every* degree ℓ ; any closed-form evaluation of its field therefore needs the loop's Legendre coefficients *and* a truncation of an infinite series, mixing two independent sources of numerical error together with whatever a code being validated contributes. Instead, place the source on a spherical shell of radius c (with $c > a$, exterior to the Earth) carrying an azimuthal surface current

$$K_\phi(\theta) = K_0 S_\ell(\theta), \quad S_\ell(\theta) \equiv \sin \theta P'_\ell(\cos \theta). \quad (7)$$

This excites *exactly* one degree ℓ . The resulting field is given by a closed-form rational expression in a, c, μ_0, μ_1 (and, in the diffusive case, spherical Bessel functions at a single argument) — nothing is truncated, so any residual mismatch with a numerical solver is attributable to the solver alone.

5.2 Field representation

For an axisymmetric field with only a ϕ -component of vector potential ($B_\phi = 0$), introduce the flux function $\psi(r, \theta) = r \sin \theta A_\phi(r, \theta)$, related to $\mathbf{B}$ by

$$B_r = \frac{1}{r^2 \sin \theta} \frac{\partial \psi}{\partial \theta}, \quad B_\theta = -\frac{1}{r \sin \theta} \frac{\partial \psi}{\partial r}, \quad (8)$$

which satisfies $\nabla \cdot \mathbf{B} = 0$ identically. Separating $\psi(r, \theta) = R(r) S_\ell(\theta)$, the angular function (7) is exactly the eigenfunction

$$\sin \theta \frac{d}{d\theta} \left[\frac{1}{\sin \theta} \frac{dS_\ell}{d\theta} \right] = -\ell(\ell+1) S_\ell(\theta) \quad (9)$$

(verified symbolically for $\ell = 1, 2, 3$; holds for all integer $\ell \geq 1$ by the Legendre ODE), so the radial equation in a region of conductivity σ , permeability μ is

$$R''(r) - \frac{\ell(\ell+1)}{r^2} R(r) + k^2 R(r) = 0, \quad k^2 = -i\omega\mu\sigma \quad (10)$$

(the sign fixed as in Remark 1; $k = 0$ in vacuum). The general solution of (10) is

$$R(r) = \begin{cases} \alpha r j_\ell(kr) + \beta r y_\ell(kr), & k \neq 0, \\ \alpha r^{\ell+1} + \beta r^{-\ell}, & k = 0, \end{cases} \quad (11)$$

both branches verified symbolically ($r j_\ell(kr)$ satisfies (10) exactly for $\ell = 0, 1, 2, 3$; the $k \rightarrow 0$ limit reproduces the power-law solutions, as it must).

5.3 Three regions

- **Region 1** ($0 < r < a$, conductor σ, μ_1): regularity at the origin excludes y_ℓ ,

$$R_1(r) = C r j_\ell(k_1 r), \quad k_1^2 = -i\omega\mu_1\sigma.$$

- **Region 2** ($a < r < c$, vacuum, μ_0): $k = 0$,

$$R_2(r) = A r^{\ell+1} + B r^{-\ell}.$$

- **Region 3** ($r > c$, vacuum, μ_0): regularity at infinity excludes the growing term,

$$R_3(r) = D r^{-\ell}.$$

5.4 Matching conditions

Continuity of ψ (normal B) at both interfaces, and continuity of tangential $H = B/\mu$ at $r = a$ (source-free) together with its prescribed jump at $r = c$ (surface current (7), from $\hat{r} \times (\mathbf{H}_{\text{out}} - \mathbf{H}_{\text{in}}) = \mathbf{K}$):

$$R_1(a) = R_2(a), \quad \frac{R'_1(a)}{\mu_1} = \frac{R'_2(a)}{\mu_0}, \quad (12)$$

$$R_2(c) = R_3(c), \quad \frac{R'_2(c)}{\mu_0} - \frac{R'_3(c)}{\mu_0} = K_0. \quad (13)$$

This is a 4×4 linear system for (A, B, C, D) , solved symbolically below and cross-checked against a known limit.

5.5 Magnetostatic case ($\omega = 0$)

Here $R_1 = C r^{\ell+1}$ (no Bessel functions needed). Solving (12)–(13) gives, with

$$\Delta \equiv \ell(\mu_0 + \mu_1) + \mu_0, \quad (14)$$

the closed form

$$A = \frac{K_0 \mu_0}{2\ell + 1} c^{-\ell}, \quad (15)$$

$$C = \frac{K_0 \mu_0 \mu_1}{\Delta} c^{-\ell}, \quad (16)$$

$$B = \frac{K_0 \mu_0 (\ell + 1) (\mu_1 - \mu_0)}{(2\ell + 1) \Delta} a^{2\ell+1} c^{-\ell}, \quad (17)$$

$$D = A c^{2\ell+1} + B. \quad (18)$$

(All four verified by direct symbolic solution of the 4×4 system and sanity-checked: setting $\mu_1 = \mu_0$ collapses the sphere's presence entirely, giving $B = 0$ and $A = C$, exactly as required when there is no permeability contrast to scatter off.)

The fields follow from $H = B_{\text{field}}/\mu$, $H_r = R(r)S'_\ell(\theta)/(\mu r^2 \sin \theta)$, $H_\theta = -R'(r)S_\ell(\theta)/(\mu r \sin \theta)$, with R, μ taken from the appropriate region.

5.6 Diffusive case (general ω)

With $R_1(r) = C r j_\ell(k_1 r)$, $k_1^2 = -i\omega\mu_1\sigma$, define

$$F_a \equiv a j_\ell(k_1 a), \quad F'_a \equiv \left. \frac{d}{dr} [r j_\ell(k_1 r)] \right|_{r=a} = j_\ell(k_1 a) + k_1 a j'_\ell(k_1 a)$$

(both directly computable, e.g. `scipy.special.spherical_jn(1, x)` and its `derivative=True` option). Solving (12)–(13):

$$A = \frac{K_0 \mu_0}{2\ell + 1} c^{-\ell}, \quad (19)$$

$$C = \frac{K_0 \mu_0 \mu_1}{\ell \mu_1 F_a + \mu_0 a F'_a} a^{\ell+1} c^{-\ell}, \quad (20)$$

$$B = \frac{K_0 \mu_0 [(\ell + 1) \mu_1 F_a - \mu_0 a F'_a]}{(2\ell + 1) [\ell \mu_1 F_a + \mu_0 a F'_a]} a^{2\ell+1} c^{-\ell}, \quad (21)$$

$$D = A c^{2\ell+1} + B. \quad (22)$$

(Structurally identical to the magnetostatic case with $F_a \rightarrow a^{\ell+1}$, $F'_a \rightarrow (\ell+1)a^\ell$ recovering Δ above — this is itself a useful internal consistency check to run in code: evaluate the diffusive formulas at $\omega \rightarrow 0^+$ and confirm they converge to the magnetostatic ones.)

Inside the conductor the induced electric field follows directly from $E_\phi = -i\omega A_\phi = -i\omega \psi/(r \sin \theta)$:

$$E_\phi(r, \theta) = -i\omega C j_\ell(k_1 r) \frac{S_\ell(\theta)}{\sin \theta}, \quad 0 < r < a. \quad (23)$$

5.7 Recipe for benchmarking a numerical solver

1. Fix $\ell \geq 1$, $a, c > a$, $\sigma, \mu_1, \mu_0, \omega$ (or $\omega = 0$), K_0 .
2. Impose the surface current (7) at $r = c$ as the numerical solver's source (this is the only place the choice of ℓ enters the source term).
3. Evaluate A, B, C, D above (magnetostatic or diffusive as appropriate) and reconstruct H_r, H_θ, E_ϕ on a grid.
4. Compare directly against the numerical solver's output at the same (r, θ) points; since nothing here is truncated or approximated, any discrepancy is attributable to the numerical scheme (discretization, absorbing boundary, mesh resolution near $r = a, c$, etc.), not to the reference solution.
5. Sweep $\ell = 1, 2, 3, \dots$ and $\omega \rightarrow 0$ vs. ω finite as independent checks — a solver that matches well at $\omega = 0$ but drifts at finite ω likely has a phase-convention bug of exactly the kind in Remark 1.